

#14Days14Chapters

DAY 8

[Aim: 100/100 in Maths]

PROBABILITY

#Random Experiment: An experiment in which all the possible outcomes are known and the exact outcome cannot be predicted in advance is called a random experiment.

Example:

- ✓ i. Tossing of a coin.
- ✓ ii. Rolling of a die dice.
- ✓ iii. Drawing a card from a pack of well shuffled cards.

#Sample Space: The sample space of an experiment is the set of all possible outcomes. (S)

Example:

i. A coin is tossed

$$S = \{H, T\} = 2$$

ii. Two coins are tossed simultaneously
(एक साथ)

$$S = \left\{ \begin{array}{cc} HH \\ TT \\ TH \\ HT \end{array} \right\} = 4$$

iii. A die is rolled

$$S = \{1, 2, 3, 4, 5, 6\} = 6$$

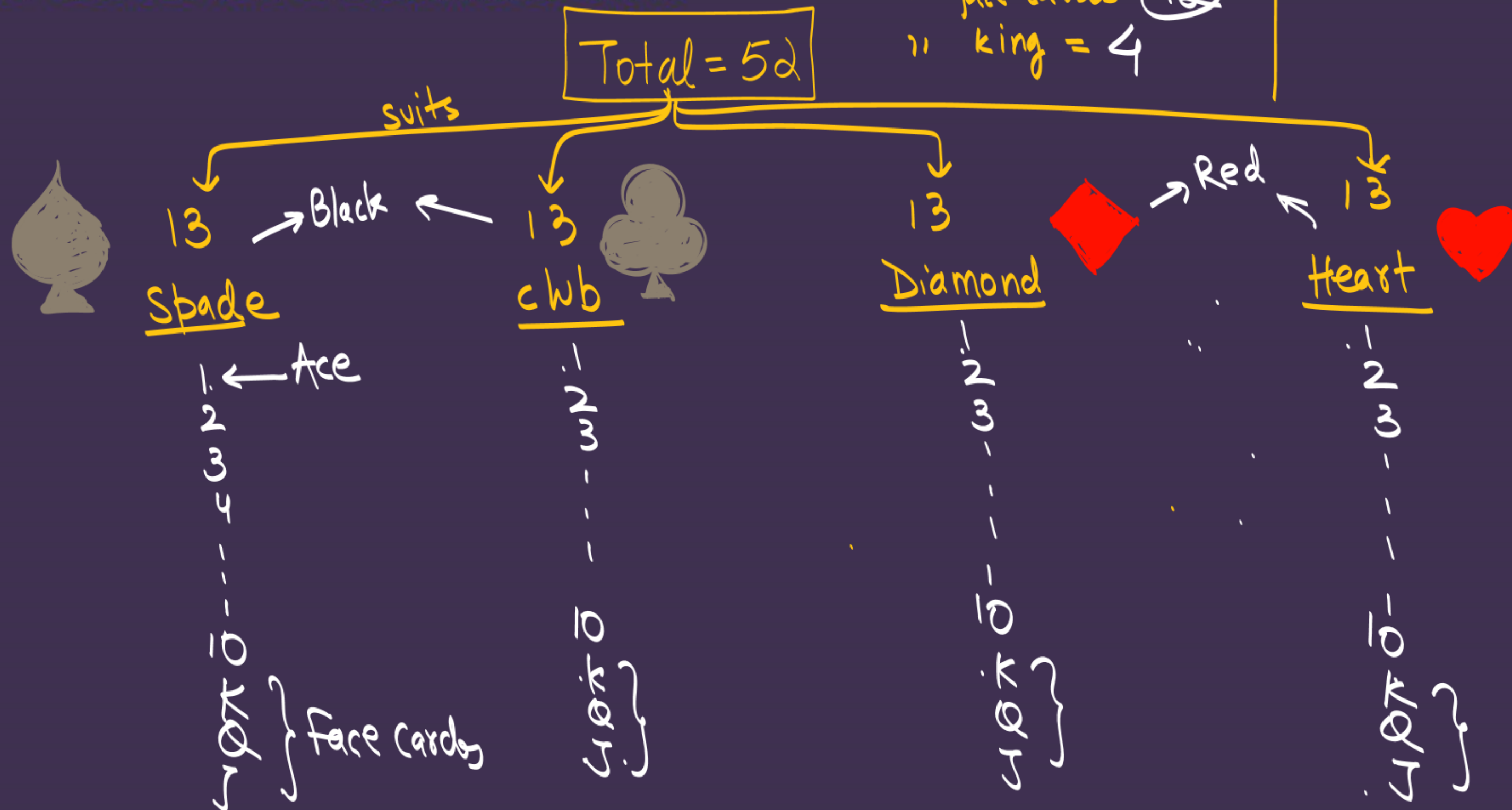
iv. Two dice are rolled simultaneously

$$\begin{aligned}
 S = \{ & (1\ 1), (1\ 2), (1\ 3), (1\ 4), (1\ 5), (1\ 6) \\
 & (2\ 1), (2\ 2), (2\ 3), (2\ 4), (2\ 5), (2\ 6) \\
 & (3\ 1), (3\ 2), (3\ 3), (3\ 4), (3\ 5), (3\ 6) \\
 & (4\ 1), (4\ 2), (4\ 3), (4\ 4), (4\ 5), (4\ 6) \\
 & (5\ 1), (5\ 2), (5\ 3), (5\ 4), (5\ 5), (5\ 6) \\
 & (6\ 1), (6\ 2), (6\ 3), (6\ 4), (6\ 5), (6\ 6) \}
 \end{aligned}$$

$$S = 36$$

#Patto ki Jaankaari:

Total black cards = $13 + 13 = 26$ total ace = 4
" Red cards = $13 + 13 = 26$
" face cards = 12
" king = 4



^(E)
#Event: The collection of specific outcomes from all possible outcomes is called an Event.

Example:

Two coins are tossed, write an event at least one tail comes.

$$S = \text{all possible outcomes} \Rightarrow \{HH, HT, TH, TT\} = 4$$

$$E = \text{at least one tail} \Rightarrow \{HT, TH, TT\} = 3$$

Probability = $\frac{\text{favourable outcome}}{\text{all possible outcome}}$

 $\Rightarrow \frac{3}{4}$

#LP: One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting: $S = 52$

(i) A king of the red suit

$$P_1 = \frac{2}{52}$$

(ii) A face card

$$\longrightarrow P_2 = \frac{12}{52}$$

(iii) A red face card

$$\longrightarrow P_3 = \frac{6}{52}$$

(iv) A queen of the black suit

$$\longrightarrow P_4 = \frac{2}{52}$$

(v) A jack of hearts

$$\longrightarrow P_5 = \frac{1}{52}$$

(vi) A spade

$$\longrightarrow P_6 = \frac{13}{52} \Rightarrow \frac{1}{4}$$

#LP: From a deck of 52 playing cards, Jack and King of red colour and Queen and Aces of black colour are removed. The remaining cards are mixed and a card is drawn at random. Find the probability that the drawn card is:

i. A black Queen $P_1 = \frac{0}{44} = 0$

✓ ii. A card of red colour $\rightarrow P_2 = \frac{\text{red colour}}{44}$
 $= \frac{26-4}{44} \Rightarrow \frac{22}{44} \Rightarrow \frac{1}{2}$

✓ iii. A jack of black colour $\rightarrow P_3 = \frac{\text{black Jack}}{44} \Rightarrow \frac{2}{44}$

✓ iv. A face card $\rightarrow P_4 = \frac{\text{face card}}{44} \Rightarrow \frac{12-6}{44} \Rightarrow \frac{6}{44}$

$$S = 52 - 4 - 4$$

↑ ↑
J & K red Q & A black

$$S = 44$$

#LP : Two dice are rolled, write the event of the following conditions :

→ $S = 36$

i. Sum of the numbers of two dice are 5

$$P_1 = \frac{\text{sum } 5}{36} = \frac{4}{36} \Rightarrow \left(\frac{1}{9}\right)$$

ii. Difference of the numbers on the two dice are 2

$$P_2 = \frac{\text{diff. } 2}{36} \Rightarrow \left(\frac{8}{36}\right)$$

iii. Same number on both dice

$$P_3 = \frac{\text{same no.}}{36} \Rightarrow \left(\frac{6}{36}\right) \Rightarrow \left(\frac{1}{6}\right)$$

iv. Different number on both dice

$$P_4 = \frac{36 - 6}{36} \Rightarrow \frac{30}{36} \Rightarrow \left(\frac{5}{6}\right)$$

$$1 - \frac{1}{6} \Rightarrow \left(\frac{5}{6}\right)$$

$$\begin{aligned} P(E) + P(\bar{E}) &= 1 \\ P(\text{same}) + P(\text{diff}) &= 1 \\ \frac{1}{6} + P(\text{diff}) &= 1 \\ P(\text{diff}) &= 1 - \frac{1}{6} \end{aligned}$$

$$\begin{aligned}
 S = \{ & \textcircled{(1\ 1)}, (1\ 2), \text{~}(1\ 3), \text{~}(1\ 4), (1\ 5), (1\ 6) \\
 & (2\ 1), \textcircled{(2\ 2)}, \text{~}(2\ 3), \text{~}(2\ 4), (2\ 5), (2\ 6) \\
 & \text{~}(3\ 1), \text{~}(3\ 2), \textcircled{(3\ 3)}, (3\ 4), \text{~}(3\ 5), (3\ 6) \\
 & \text{~}(4\ 1), \text{~}(4\ 2), (4\ 3), \textcircled{(4\ 4)}, (4\ 5), \text{~}(4\ 6) \\
 & (5\ 1), (5\ 2), \text{~}(5\ 3), (5\ 4), \textcircled{(5\ 5)}, (5\ 6) \\
 & (6\ 1), (6\ 2), (6\ 3), \text{~}(6\ 4), (6\ 5), \textcircled{(6\ 6)} \}
 \end{aligned}$$

$$S = 36$$

#Probability of occurrence of an event E, denoted by $P(E)$ is:

$$P(E) = \frac{\text{favourable outcome}}{\text{all possible outcome}} = \frac{E}{S}$$

- $0 \leq P(E) \leq 1$
- $P(E) + P(\bar{E}) = 1$

For a game.

$$P(\text{Winning}) = 0.5$$

$$P(\text{Losing}) = ?$$

$$P(E) + P(E') = 1$$

$$P(W) + P(L) = 1$$

$$0.5 + P(L) = 1 \rightarrow 1 - 0.5 = 0.5$$

#LP: A bag contains 4 red, 5 black and 6 white balls. A ball is drawn from the bag at random. Find the probability that the ball drawn is:

(i) White $P(W) = \frac{W}{S} = \left(\frac{6}{15}\right)$

$$\begin{array}{l} \alpha R \rightarrow 4 \\ \checkmark B \rightarrow 5 \\ \alpha W \rightarrow 6 \end{array} \left. \vphantom{\begin{array}{l} \alpha R \rightarrow 4 \\ \checkmark B \rightarrow 5 \\ \alpha W \rightarrow 6 \end{array}} \right\} \rightarrow T = \boxed{15 = S}$$

(ii) Red $P(R) = \left(\frac{4}{15}\right)$

(iii) Not black $P(\text{not black}) = \frac{R+W}{S} \Rightarrow \frac{4+6}{15} \Rightarrow \frac{10}{15} \Rightarrow \frac{2}{3} \Rightarrow \left(\frac{2}{3}\right)$

(iv) Red or White $P(R \text{ or } W) = \frac{R+W}{S} = \left(\frac{2}{3}\right)$

(v) Neither red nor white $P(B) = \frac{B}{S} = \frac{5}{15} \Rightarrow \left(\frac{1}{3}\right)$

#LP: The probability of selecting a rotten apple randomly from a heap of 900 apples is 0.18. What is the number of rotten apples in the heap? **[CBSE 2017]**

$$S = 900$$

given $\rightarrow P(\text{rotten}) = 0.18$

$$\frac{\text{no. of rotten}}{900} = \frac{0.18}{100}$$

$$\text{no. of rotten apple} \Rightarrow 18 \times 9 = \underline{\underline{162}}$$

#LP: The probability of getting a bad egg in a lot of 400 eggs is 0.045. The number of good eggs in the lot

is: (1 Mark) (CBSE 2024)

(a) 18

(b) 180

☒ (c) 382

(d) 220

$$S = 400$$

$$\text{given} \rightarrow P(\text{bad}) = 0.045$$

$$\Rightarrow \frac{\text{no. of bad eggs}}{400} = \frac{0.045}{1000}$$

$$\rightarrow \text{no. of bad eggs} = \frac{45 \times 4^2}{10^2} \Rightarrow \underline{18}$$

$$\therefore \text{good eggs} \Rightarrow \overset{\text{total}}{400} - \overset{\text{bad}}{18} \Rightarrow \underline{\underline{382}}$$

#LP: The probability of the occurrence of an event is denoted by p, and the probability of the non-occurrence of the event is denoted by q. The relation between p and q is (1 Mark) (2023)

(a) $p + q = 1$

(b) $p = 1, q = 1$

(c) $p = q - 1$

(d) $p + p + 1 = 0$

$$P(E) + P(\bar{E}) = 1$$

$$p + q = 1$$

#LP: In a group of 20 people. 5 can't swim. If one person is selected at random, then the probability that he/she can swim is (1 Mark) (2023)

☒ (a) $3/4$

(b) $1/3$

(c) 1

(d) $1/4$

$$T = \boxed{20 = S}$$

$$\text{Cannot swim} = 5$$

$$\text{Swim} = 20 - 5 = \boxed{15}$$

$$P(\text{swim}) = \frac{15}{20} \rightarrow \left(\frac{3}{4} \right)$$

#LP:

If in a lottery, there are 10 prizes and 30 blanks, then the probability of winning a prize is :

(A) $\frac{1}{4}$

(B) $\frac{1}{3}$

(C) $\frac{3}{4}$

(D) $\frac{2}{3}$

[CBSE 2025]

$$P(\text{prize}) = \frac{10}{40} \rightarrow \frac{1}{4}$$

#LP: Three coins are tossed together. Find the probability of getting:

(i) exactly two heads $P_1 = \frac{3}{8}$
(= 2)

(ii) ^(max)at most two heads $\Rightarrow \frac{7}{8}$
(≤ 2)

(iii) at least one head and one tail $\Rightarrow \frac{6}{8}$

(iv) no tails $\Rightarrow \frac{1}{8}$

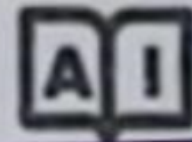
$S =$

✓	HHH	✓
✓	HHT	✓
✓	HTH	✓
✓	TTH	✓
✓	THT	✓
✓	HTT	✓
✓	THT	✓
✓	TTT	✓

8

#LP:

A game consists of tossing a coin 3 times and noting the outcome each time, if getting the same result in all the tosses is a success, find the probability of losing the game.



[Delhi Set-1, 2019]

$$P(S) + P(L) = 1$$

$$\frac{1}{4} + P(L) = 1$$

$$P(L) = 1 - \frac{1}{4}$$

$$\frac{3}{4}$$

$$S = 8$$

Success $\rightarrow$ all same

$$P(\text{Success}) = \frac{2}{8} \Rightarrow \frac{1}{4}$$

$\left\{ \begin{array}{l} HHH \\ TTT \end{array} \right\}$

$$\therefore P(\text{Losing}) = 1 - \frac{1}{4} = \frac{3}{4} \quad \checkmark \text{ Ans}$$

#LP: Five cards—the ten, jack, queen, king and ace of diamonds, are well-shuffled with their face downwards. One card is then picked up at random.

(i) What is the probability that the card is the queen?

(ii) If the queen is drawn and put aside, what is the probability that the second card picked up is (a) an ace? (b) a queen?

$$S = \{10, J, Q, K, A\} = 5$$

$$(i) P(\text{Queen}) = \frac{1}{5}$$

$$(ii) \text{ New } S = \{10, J, K, A\} = 4$$

$$(a) P(\text{ace}) = \frac{1}{4} \quad (b) P(Q) = \frac{0}{4} = 0 //$$

#LP:

A bag contains balls numbered 2 to 91 such that each ball bears a different number. A ball is drawn at random from the bag. Find the probability that

[CBSE 2025]

- (i) it bears a 2-digit number
- (ii) it bears a multiple of 1.

$$S = \{2, 3, 4, 5, \dots, 91\} = 90$$

$$(i) P(\text{two digit no}) = \frac{82}{90} \checkmark$$

$\{10, 11, 12, \dots, 91\}$

$$(ii) P(\text{mult of 1}) = \frac{90}{90} = \underline{\underline{1}} \checkmark$$

$\{2, 3, 4, \dots, 91\}$

#LP:

The number of red balls in a bag is 10 more than the number of black balls. If the probability of drawing a red ball at random from this bag is $\frac{3}{5}$, then the total number of balls in the bag is: 1

(a) 50

(b) 60

(c) 80

(d) 40

[CBSE 2025]

$$R = 10 + B$$

$$\text{Total} = R + B \rightarrow 30 + 20 = 50$$

$$P(R) = \frac{3}{5}$$
$$\frac{R}{R+B} = \frac{3}{5}$$

$$5R = 3R + 3B$$

$$2R = 3B$$

$$2(10+B) = 3B$$
$$20 + 2B = 3B$$

$$20 = B$$

$$R = 10 + B = 10 + 20 = 30$$

#LP:
V. msp

Two friends Anil and Ashraf were born in the December month in the year 2010. Find the probability that :

[CBSE 2025] 2

- (i) they share same date of birth.
(ii) they have different dates of birth.

$S = 31 \times 31$

(i) $P(\text{same DOB}) = \frac{31}{31 \times 31}$

$\frac{1}{31}$

(ii) $P(\text{dif}) = 1 - \frac{1}{31} = \frac{30}{31}$

- $S = \{$
- | | | | | | |
|--------------|--------------|--------------|-----------|-----------|------------------|
| <u>(1,1)</u> | (1,2) | (1,3) | (1,4) | - - - - - | (1,31) → 31 |
| (2,1) | <u>(2,2)</u> | (2,3) | (2,4) | - - - - - | (2,31) → 31 |
| (3,1) | (3,2) | <u>(3,3)</u> | (3,4) | - - - - - | (3,31) → 31 |
| (4,1) | (4,2) | <u>(4,3)</u> | - - - - - | - - - - - | (4,31) |
| - - - - - | - - - - - | - - - - - | - - - - - | - - - - - | - - - - - |
| (31,1) | (31,2) | (31,3) | - - - - - | - - - - - | (<u>31,31</u>) |
- $\downarrow$ 31

#LP: Two customers Shyam and Ekta are visiting a particular shop in the same week (Tuesday to Saturday). Each is equally likely to visit the shop on any day as on another day. What is the probability that both will visit the shop on (i) the same day? (ii) consecutive days? (iii) different days? **Exemplar**

$$S = \{ \underline{(T, T)} \quad \underline{(T, W)} \quad (T, Th) \quad (T, F) \quad (T, S) \}$$

$$(i) P(\text{same}) = \frac{8}{5 \times 5} \times \left(\frac{1}{5} \right) \quad \underline{(W, T)} \quad \underline{(W, W)} \quad \underline{(W, Th)} \quad (W, F) \quad (W, S)$$

$$S = 5 \times 5$$

$$(ii) P(\text{diff}) = 1 - \frac{1}{5} \rightarrow \left(\frac{4}{5} \right) \quad (Th, T) \quad \underline{(Th, W)} \quad \underline{(Th, Th)} \quad \underline{(Th, F)} \quad (Th, S)$$

$$(iii) P(\text{consec}) = \frac{8}{25} \quad \begin{array}{l} (F, T) \quad (F, W) \quad \underline{(F, Th)} \quad \underline{(F, F)} \quad \underline{(F, S)} \\ (S, T) \quad (S, W) \quad (S, Th) \quad \underline{(S, F)} \quad \underline{(S, S)} \end{array} \}$$

#LP:

If a number x is chosen at random from the numbers $\{-3, -2, -1, 0, 1, 2, 3\}$, then find the probability of $x^2 < 4$. [CBSE 2020]

$$x = \{-3, -2, -1, 0, 1, 2, 3\} \Rightarrow S = 7$$

$$x^2 = \{9, 4, 1, 0, 1, 4, 9\}$$

$$P(x^2 < 4) = \frac{3}{7}$$

#LP: What is the probability that a non-leap year has 53 Tuesdays?

Non-leap year $\Rightarrow$ 52 weeks + 1 day

52 tuesday
confirm

$$7 \overline{) 365} \begin{array}{r} 52 \\ 35 \\ \hline 15 \\ 14 \\ \hline 1 \end{array}$$

all possibilities (s) = {s, m, t, w, th, f, sat} = 7

$$P(\text{tuesday}) = \frac{1}{7}$$

#LP: What is the probability that a leap year has 53 Tuesdays?

leap year $\Rightarrow$ 52 week + 2 days

$\uparrow$
52 tues confirm.

$$7 \overline{) 366} \begin{array}{r} 52 \\ 35 \\ \hline 16 \\ 14 \\ \hline 2 \end{array}$$

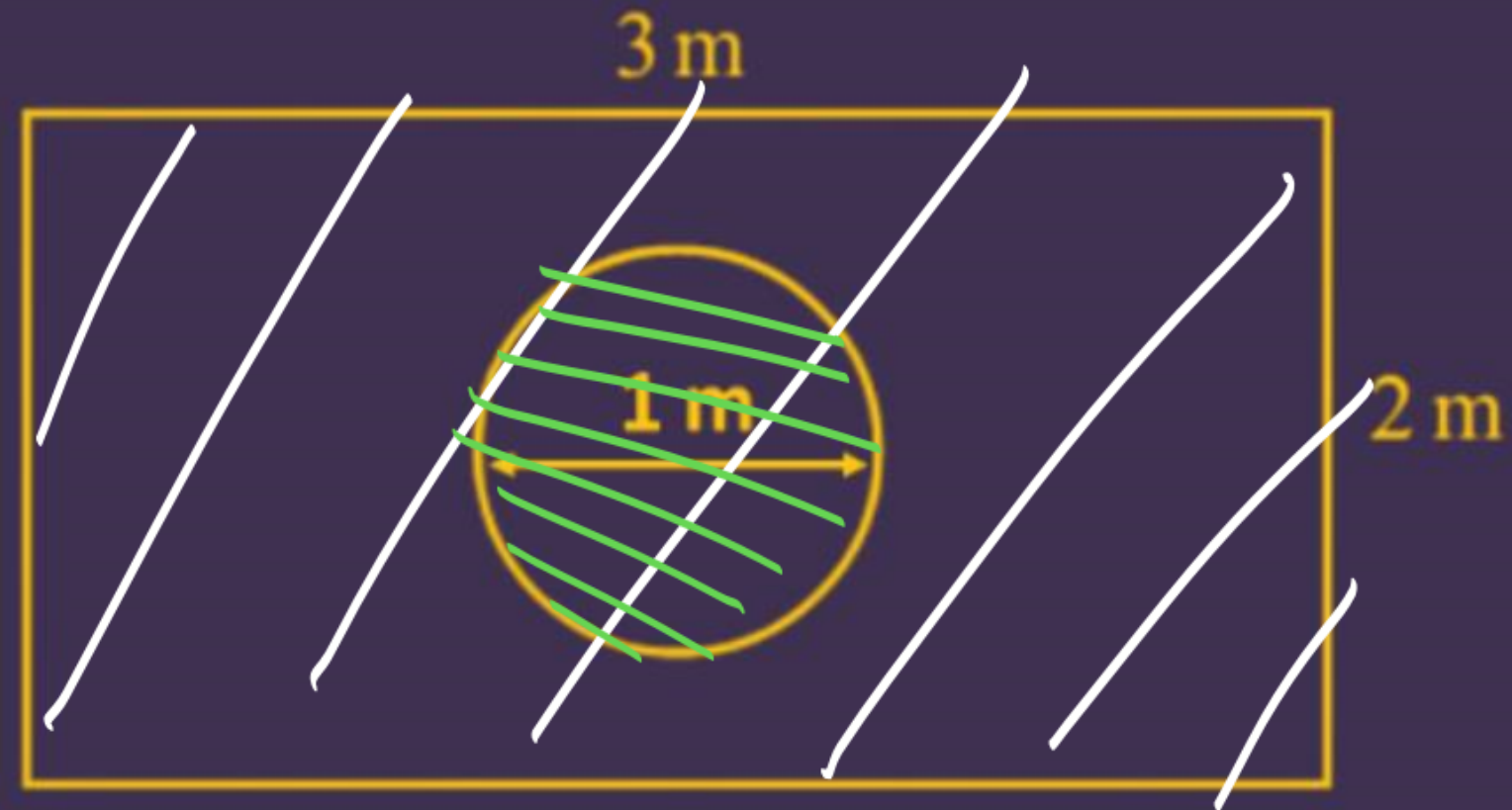
(S) possibilities = $\{ \text{SM}, \underline{\text{MT}}, \underline{\text{TW}}, \text{WTh}, \text{ThF}, \text{FSat}, \text{SatS} \} = 7$

$$P(\text{at least 1 tuesday}) = \frac{2}{7}$$

#LP: Suppose you drop a die at random on the rectangular region shown in fig. below. What is the probability that it will land inside the circle with diameter 1 m?

$$S = \text{Rectangle} \Rightarrow l \times b \\ \Rightarrow 3 \times 2 \Rightarrow 6\text{m}^2$$

$$P(\text{Circle}) = \frac{\pi r^2}{6\text{m}^2} \\ \Rightarrow \left[\frac{\frac{22}{7} \times \left(\frac{1}{2}\right)^2}{6} \right]$$



Peter throws two different dice together and finds the product of the two numbers obtained. Rina throws a die and squares the number obtained. Who has the better chance to get the number 25? (CBSE 2017)

Peter 2 dice $\rightarrow$ Product

$$S = 36$$

$$P(\text{Product } 25) = \frac{1}{36}$$

Rina 1 die

$$S = \{1, 2, 3, 4, 5, 6\}$$

1 4 9 16 25 36

$$P(\text{square } 25) = \frac{1}{6}$$

$$\frac{1}{36}$$

<

$$\frac{1}{6}$$



Reena Joshi

$$\begin{aligned}
 S = \{ & (1\ 1), (1\ 2), (1\ 3), (1\ 4), (1\ 5), (1\ 6) \\
 & (2\ 1), (2\ 2), (2\ 3), (2\ 4), (2\ 5), (2\ 6) \\
 & (3\ 1), (3\ 2), (3\ 3), (3\ 4), (3\ 5), (3\ 6) \\
 & (4\ 1), (4\ 2), (4\ 3), (4\ 4), (4\ 5), (4\ 6) \\
 & (5\ 1), (5\ 2), (5\ 3), (5\ 4), (5\ 5), (5\ 6) \\
 & (6\ 1), (6\ 2), (6\ 3), (6\ 4), (6\ 5), (6\ 6) \}
 \end{aligned}$$

$$S = 36$$

#Sure event: An event that is certain to happen is called sure event.

Example:

A dice is thrown, find the probability of getting a number less than 7.

$$P(\text{sure event}) = 1$$

#Impossible Event: An event that cannot happen is impossible event.

Example:

A dice is thrown, find the probability of getting a number more than 6.

$$P(\text{impossible event}) = 0$$

#LP: Directions: In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:

- ☒ (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
- ☐ (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
- ☐ (c) Assertion (A) is true but reason (R) is false.
- ☐ (d) Assertion (A) is false but reason (R) is true. ✓

#LP: Assertion (A): The probability of selecting a number at random from the numbers (1 to 20) is 1.

Reason (R) : For any event E, if $P(E) = 1$, then E is called a sure event.

$$1 \longrightarrow 20$$
$$P(\text{any number}) = \frac{1}{20} \quad \text{False}$$

#LP:

Q4: Assertion (A): In a cricket match, a batsman hits a boundary 9 times out of 45 balls he plays. The probability that in a given ball, he does not hit the boundary is $4/5$. 7

Reason (R): $P(E) + P(\text{not } E) = 1$. (1 Mark) (CBSE 2024) 7

$$S = 45$$

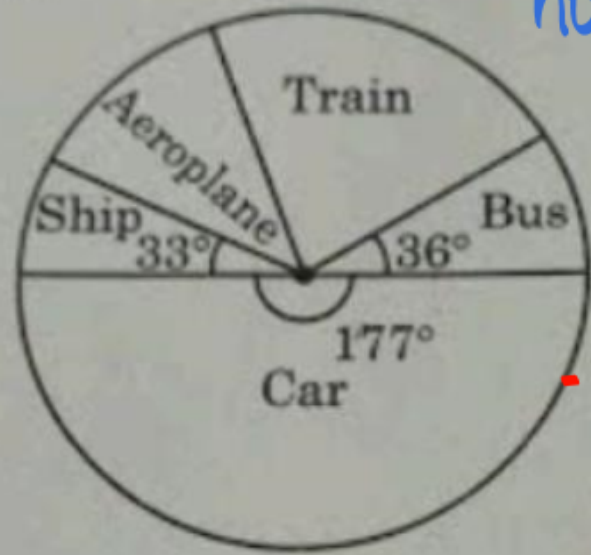
$$\text{boundary} = 9$$

$$P(b) = \frac{9}{45} \Rightarrow \frac{1}{5}$$

$$P(b') = 1 - \frac{1}{5} \Rightarrow \frac{4}{5}$$

(A) ✓

38. In a survey on holidays, 120 people were asked to state which type of transport they used on their last holiday. The following pie chart shows the results of the survey.



no. of people car $\Rightarrow \frac{177}{360} \times 120 = 59$

$N(\text{Bus}) = \frac{36}{360} \times 120 = 12$

$N(\text{Ship}) = \frac{33}{360} \times 120 = 11$

Observe the pie chart and answer the following questions :

- (i) If one person is selected at random, find the probability that he/she travelled by bus or ship.

$P(\text{Bus or Ship}) \Rightarrow \frac{12+11}{120} \Rightarrow \frac{23}{120}$

- (ii) Which is most favourite mode of transport and how many people used it? $\rightarrow \text{Car} = 59$

- (iii) (a) A person is selected at random. If the probability that he did not use train is $\frac{4}{5}$, find the number of people who used train.

OR

- (iii) (b) The probability that randomly selected person used aeroplane is $\frac{7}{60}$. Find the revenue collected by air company at the rate of ₹ 5,000 per person.

$P(T') = \frac{4}{5} \quad | \quad N(T) = ?$

$P(T) = 1 - \frac{4}{5}$

$P(T) = \frac{1}{5}$

$\frac{N(T)}{120} = \frac{1}{5}$

$N(T) = 24$

(b) $P(A) = \frac{7}{60}$

$\frac{N(A)}{120} = \frac{7}{60} \Rightarrow N(A) = 14$

Rev $\Rightarrow ₹ 14 \times 5000 \rightarrow ₹ 70000/-$

8/14 Completed!

Thank you Goodies !!

Kal → 7 PM

